

Manuscript-ready results

Confidence-Driven Occlusion-Aware Camera-Radar Fusion & Hidden Hazard Prediction for ADAS - recomputed from raw data by scripts/audit/

Every figure below was recomputed from raw data by the audit scripts in `scripts/audit/`. Source file and sample count are given for each block so each number can be traced. **Numbers not listed here were not computed and must not be claimed.**

Dataset

The evaluation dataset comprises **15,270 frames** across **170 episodes** in a single CARLA town (`Town10HD_0pt`), containing **55,803 object instances** of which **18,096** are vulnerable road users. Per-object occlusion tiers: OCCLUDED 20,912, PARTIAL 5,308, VISIBLE 29,583. All frames are clear-daylight; CARLA 0.10.0's weather API is non-functional on this build, so adverse conditions are evaluated by photometric degradation of recorded frames rather than by simulated weather.

Source: `results/audit/dataset_statistics.json`.

Splits

Crops are split **by episode, never by frame**: 144 training and 26 validation episodes (12,552 validation crops). Episode disjointness was verified programmatically (PASS – no episode appears on both sides). **No independent test split exists**: the best-epoch checkpoint is selected on the same validation split on which accuracy is reported.

Source: `results/audit/splits.json`.

Evidential classifier

On the 12,552-crop episode-disjoint validation split, the evidential classifier attains **96.79% accuracy** (macro-F1 0.965, weighted-F1 0.968). Per class: vehicle P 0.988 / R 0.949 / F1 0.968 (n=5,654), pedestrian P 0.935 / R 0.981 / F1 0.957 (n=1,326), background P 0.957 / R 0.984 / F1 0.970 (n=5,572).

Uncertainty quality, computed from raw Dirichlet outputs: **ECE 0.0622**, Brier 0.0613, NLL 0.1520, and — the operative claim — **error-detection AUROC 0.942**, i.e. predicted uncertainty separates misclassified from correctly classified crops far above chance. Uncertainty responds to occlusion as designed: partially-occluded crops carry mean uncertainty 0.216 at 88.6% accuracy (n=1,269), versus 0.133 at 97.1% for fully visible crops (n=5,711).

Caveat that must accompany this block. Fully-occluded objects are absent from the classifier's data by construction (`labels/` carries only camera-observable actors), so this comparison spans VISIBLE vs PARTIAL only.

Source: `results/audit/classification_metrics.json`, `uncertainty_metrics.json`.

Occlusion detector

At the fixed operating point ($q=0.01$, $\tau=0.001$), evaluated on **300 frames disjoint from the sample used to select that operating point** (120,000 BEV cells), the detector attains precision **0.706**, recall **0.918**, F1 **0.798**, IoU 0.664, cell agreement 0.769. Per-frame F1 is 0.778 (95% CI [0.760, 0.795], frame-level bootstrap).

The previously reported figures (0.727/0.917/0.811) come from the tuning sample itself; the optimism gap is **+0.0129 F1**, with recall essentially unchanged (-0.0004).

Performance is strongly severity-dependent: at 0.20-0.50 scene occlusion, P 0.545 / R 0.929 (n=127 frames); at 0.50-0.65 scene occlusion, P 0.806 / R 0.916 (n=94 frames); at 0.65-1.01 scene occlusion, P 0.944 / R 0.909 (n=69 frames). Recall is near-constant across severity while precision collapses in mostly-clear scenes, where few cells are genuinely occluded.

Source: results/audit/occlusion_metrics.json.

Sensor fusion: tracking

On **845 paired observations** (identical actors and frames tracked by all three configurations, drawn from 44 episodes):

Configuration	v_lat MAE (m/s)	95% CI	v_fwd MAE (m/s)	Position MAE (m)
camera	0.819	[0.691, 0.964]	1.048	2.327
radar	1.683	[1.567, 1.782]	0.155	2.457
fused	0.672	[0.489, 0.910]	0.199	2.032

Confidence intervals are episode-level bootstrap (10,000 resamples over 44 episodes), because consecutive frames within an episode are correlated and frame-level intervals would be far too narrow.

Paired significance (the defensible form of the fusion claim). Fusion is significantly better than a single sensor precisely on the axis that sensor is structurally blind to: v_lat fused vs radar (-1.011, p=0.0000); v_fwd fused vs camera (-0.849, p=0.0000); position fused vs radar (-0.425, p=0.0062). The following differences are **not** distinguishable from zero at this sample size: v_lat fused vs camera (-0.147, p=0.1994); v_fwd fused vs radar (+0.043, p=0.0892); position fused vs camera (-0.295, p=0.1534).

Source: results/audit/tracking_metrics.json, statistics.json.

Crossing-intent prediction

Configuration	n	ROC-AUC	PR-AUC	Precision	Recall	F1	Balanced acc.
camera	3,665	0.789	0.804	0.833	0.654	0.733	0.755
radar	2,491	0.743	0.829	1.000	0.018	0.035	0.509
fused	3,065	0.873	0.866	0.818	0.896	0.855	0.825

Radar alone achieves precision 1.000 at recall 0.018 — it almost never predicts a crossing, so its precision is vacuous; F1 is the honest summary. Note the three configurations score different numbers of observations (camera 3,665, radar 2,491, fused 3,065), so these rows are not paired; coverage itself differs by configuration.

Source: results/audit/intent_metrics.json.

Fault injection

Across **10 injected fault conditions**, the self-referential health monitor detects **6**: blur, darkness, fog, lens_blocked, noise, radar_dropout. It fails to detect **4**: extrinsic_drift, radar_bias, radar_clutter, radar_clutter_onset. Of those, the script classifies radar_clutter, radar_bias, radar_clutter_onset as *silent failures* — they measurably degraded the pipeline while health remained nominal, which is the most dangerous category.

Source: results/audit/fault_metrics.json.

Runtime

Perception-module latency measured on recorded frames with CARLA **out of the loop**, on NVIDIA GeForce RTX 5060 Laptop GPU: sensor_health_update 25.8 ms, midas_depth 16.3 ms, yolov8n_detect 8.9 ms, occlusion_grid_given_disparity 0.7 ms, evidential_head_per_crop 0.6 ms, ekf_tracking_step 0.1 ms, particle_filter_step_400p 0.1 ms, radar_confidence 0.0 ms. Serial sum is **52.5 ms** (19.0 FPS upper-bound cost).

The live demo's 2–9 FPS is a *synchronous-mode simulation tick rate* that includes CARLA rendering and the server round-trip. It is not perception throughput and the two must not be reported as the same quantity.

Source: `results/audit/runtime_metrics.json`.

Experiments that were NOT performed

- **G: risk engine** — SKIPPED: no risk/TTC/hazard ground truth exists in the dataset
- **M: multi-town generalisation** — SKIPPED: excluded by explicit instruction; CARLA 0.10.0 ships only Town10HD_Opt in any case
- **Hidden-target prediction** — executed but sample-limited: only 15 scorable occlusion events across 10 episodes; no headline claim is supportable.